2025 Big 12 Football

Mathematical Team Ratings and Ranking Comparison

Project report

Abstract. This project estimates relative team strength for the 16 teams in the 2025 Big 12 using score margins from 72 conference games. A least-squares model estimates one rating per team, constrained to sum to zero. The resulting ranking is compared with the supplied official standings, and fitted score margins are visualised. The Big 12 Championship Game is held out from model fitting and used as a separate test.

Key results

Measure	Result
Teams / games used	16 teams / 72 games
In-sample MAE	9.25 points
In-sample RMSE	12.04 points
In-sample winner accuracy	59 / 72 (81.9%)
Held-out championship prediction	Texas Tech by 15.48 points
Actual championship margin	Texas Tech by 27 points

1. Objective and dataset

The objective is to estimate relative team strength from game scores rather than wins and losses alone. The workbook contains 73 games: 72 conference games marked for model fitting and one championship game marked as held out. The script checks for missing team/score values, self-matchups, negative scores and duplicate game rows. The conference schedule connects all 16 teams, so ratings can be compared on one scale.

2. Mathematical model

For each game between Team A and Team B, the model assumes rA – rB ≈ scoreA – scoreB. The 72 equations form an overdetermined linear system A r ≈ b. Since only rating differences are identified, we add the constraint that the 16 ratings sum to zero. NumPy's least-squares solver (numpy.linalg.lstsq) finds ratings that minimise the sum of squared score-margin errors. A positive rating indicates above-average estimated strength on this zero-centred scale; a negative rating indicates below-average estimated strength.

3. Model-based team ratings

Texas Tech ranks first (+25.77), followed by Utah (+17.97) and BYU (+10.29). Oklahoma State ranks last (–17.54). These are model estimates, not official rankings.

Rank	Team	Rating
1	Texas Tech Red Raiders	+25.77
2	Utah Utes	+17.97
3	BYU Cougars	+10.29
4	Arizona Wildcats	+5.67
5	Kansas State Wildcats	+3.76
6	Iowa State Cyclones	+1.33

Rank	Team	Rating
7	Cincinnati Bearcats	+0.13
8	Arizona State Sun Devils	+0.11
9	Houston Cougars	-0.90
10	TCU Horned Frogs	-1.91
11	Kansas Jayhawks	-2.46
12	Baylor Bears	-4.89
13	UCF Knights	-9.31
14	West Virginia Mountaineers	-13.95
15	Colorado Buffaloes	-14.07
16	Oklahoma State Cowboys	-17.54

4. Comparison with official standings

The model ranking is compared with the official order supplied for this analysis. Mean absolute rank difference: 1.50 places. Spearman rank correlation: 0.888. Largest absolute rank difference: 5 places. Rank difference is model rank minus official rank: positive values mean the model places a team lower.

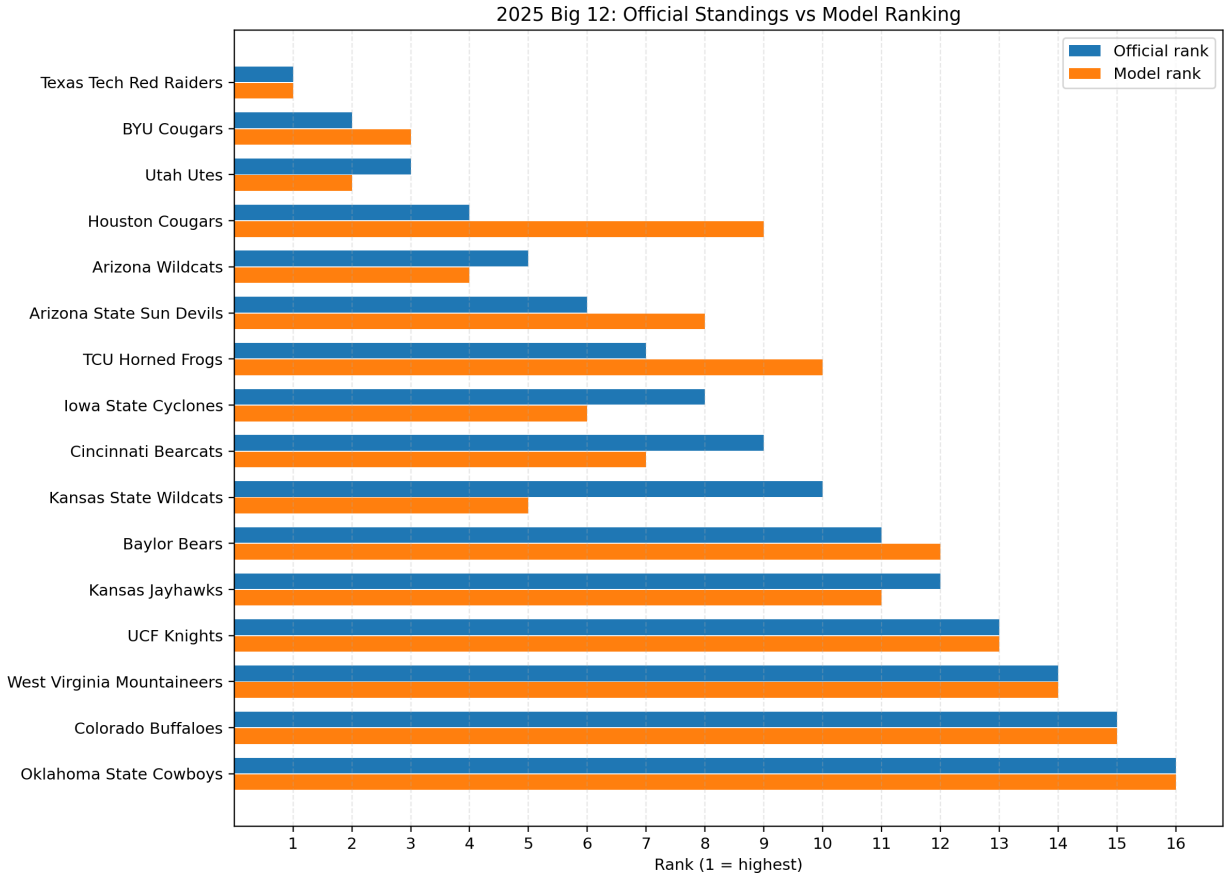


Figure 1. Side-by-side comparison of official and model positions.

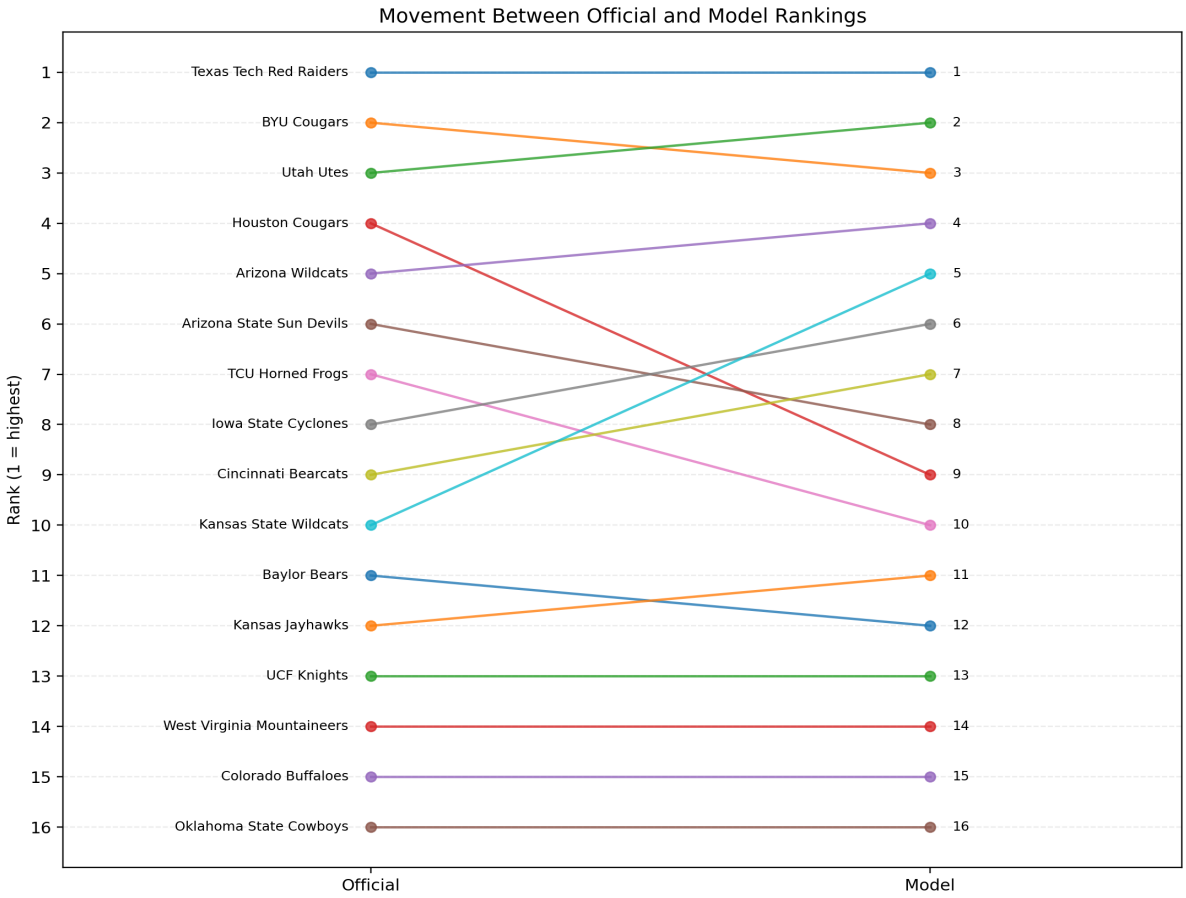


Figure 2. Each team's movement between rankings.

5. Score-margin fit

Each point compares the model-predicted margin with the actual margin. Points on the dashed diagonal are exact predictions. Points above the diagonal indicate an overestimate of Team A's margin; points below indicate an underestimate.

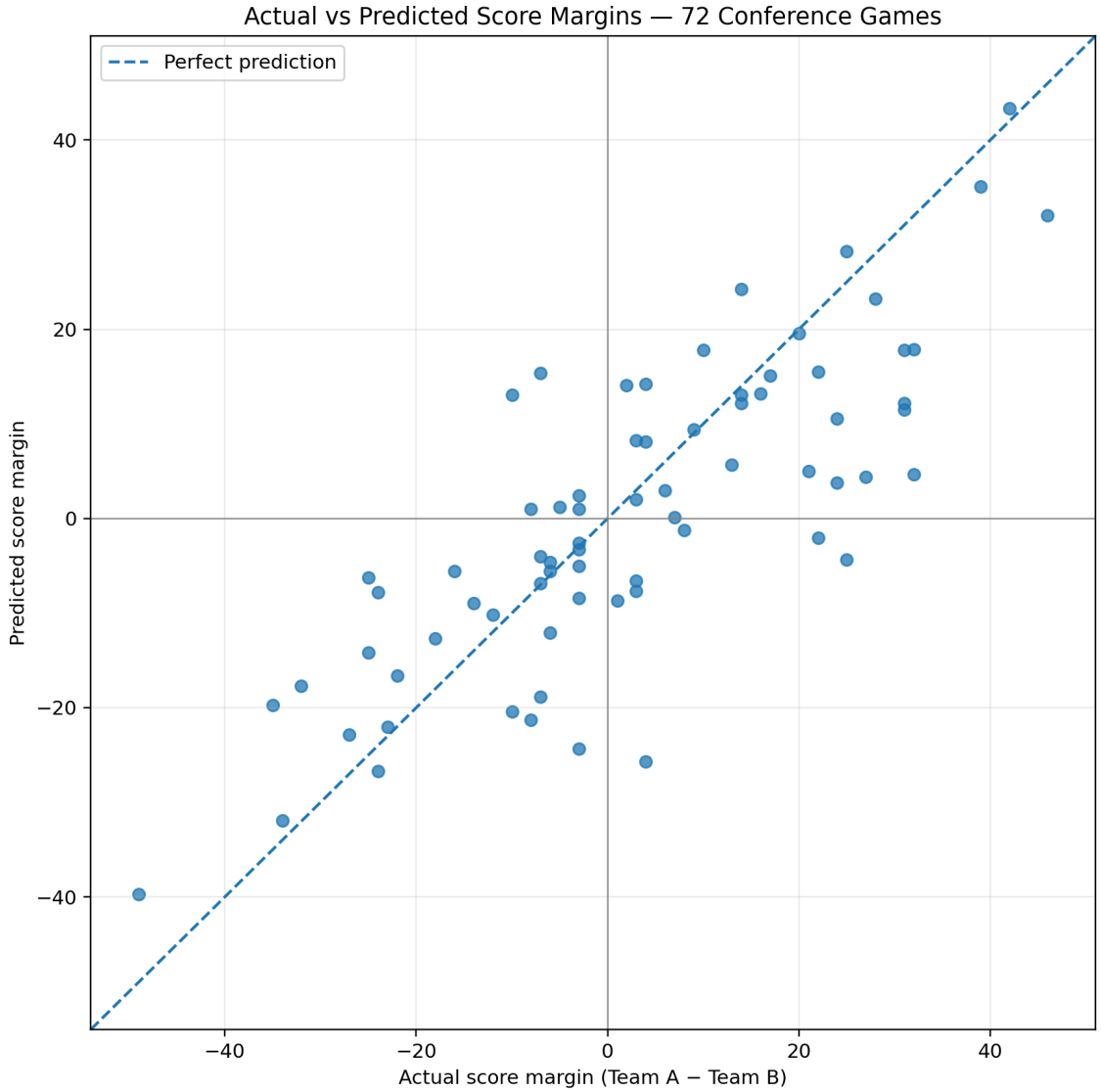


Figure 3. Actual versus predicted score margins for the 72 fitted conference games.

The in-sample MAE is 9.25 points and RMSE is 12.04 points. The model picks the correct winner in 59 of 72 games (81.9%). These are in-sample measures because the same games were used to estimate ratings and calculate the metrics; they may be optimistic.

6. Held-out championship game

The championship game was excluded from model fitting. BYU lost 7–34 to Texas Tech. The model predicted a 15.48-point advantage for Texas Tech; the actual margin was 27 points. The model selected the correct winner but underestimated the margin by 11.52 points.

7. Limitations

- Home-field advantage is not modelled.
- Injuries, player availability and roster strength are not included.
- Schedule strength is accounted for indirectly through opponents and their estimated ratings.
- Squared-error fitting gives larger misses more influence.
- The ranking comparison depends on the official standings order supplied for this analysis.
- The 72-game fit metrics are in-sample; the single championship test is too small to establish general predictive performance.

8. Conclusion

The least-squares method turns 72 conference scorelines into a consistent set of relative team ratings. Texas Tech, Utah and BYU lead the model ranking, while Oklahoma State ranks last. The model correctly identifies 81.9% of winners in the fitted games, with an average absolute margin error of 9.25 points. On the held-out championship game, it correctly predicts Texas Tech but underestimates the winning margin. This demonstrates the usefulness of score-based mathematical modelling as a complementary view of team strength, while showing why additional features and more unseen-game testing would be needed for a stronger predictive system.

Reproducibility

Run from the repository root: `pip install -r requirements.txt`, then `python src/big12_ranking.py`. The script regenerates team ratings, game residuals, metrics and the model rating chart. The comparison figures and report are included as generated project outputs.